

Spatio-temporal analysis of signals to detect drift in air quality sensors

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Highlights

Challenge

- Detecting drift in sensors
- Definition of drift : Drift is an abnormal behaviour that deviate at a time t spatially and temporally
- Methodology : Using Graph Fourier Transform and Spectrogram construction to detect drift
 - Part 1 : Test Simulated Data
 - Part 2 : Test on Real Data

Definitions

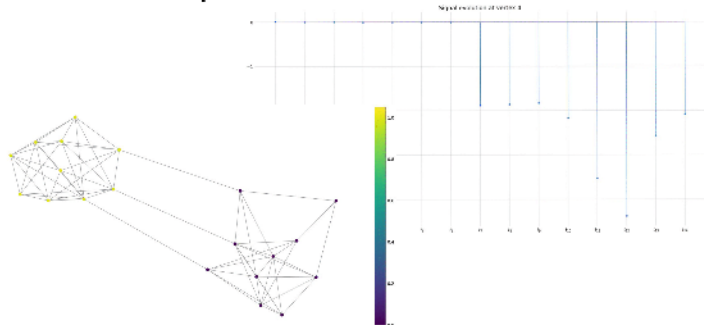
- Graph Fourier Transform (GFT):
Let $G=(V,E,W)$ a graph with :
 V : set of vertices, E : set of edges
 W : Weight matrix of G
 D : Diagonal values of W
If $L = D - W$ and U the eigenvectors of L ,
The GFT is defines as $GFT(f) = U^L f$
- Spectrogram :
Visual representation of the spectrum of frequencies of a signal as it varies with time

Simulated Drift Implementation

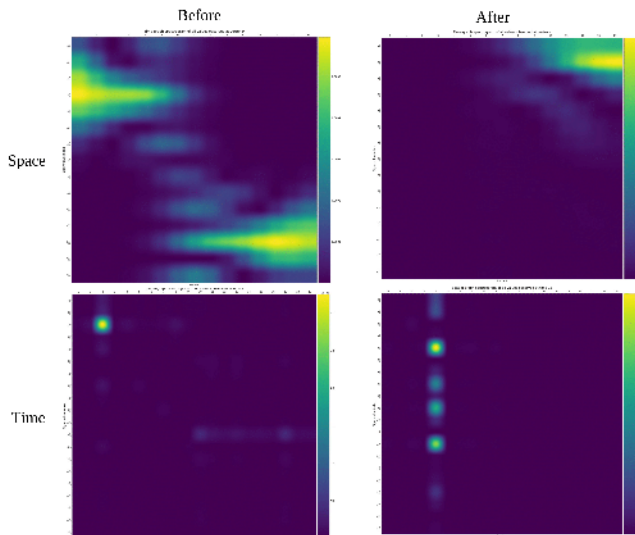
Simulation :

- Sinusoidal signal for temporal variation
- Method for spacial variation: 2 groups of signal created using 2 different eigenvectors
- Drift sensor is at vertex 0 : Wiener Process

Network in time and space

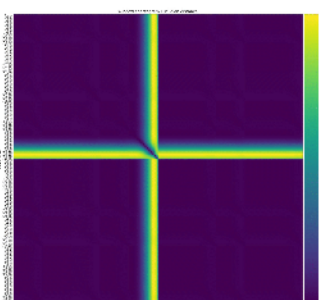


Spectrogram for disjoint GFT in time and space before and after the drift



Drift Detection

- Technique to detect outliers : Subtraction of sub-spectrograms centered
 - Find the most changing sub-spectrogram.



Data Processing

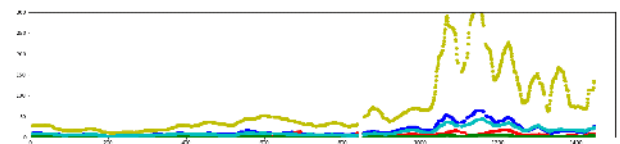
In order to obtain a good representation in the spectrogram is necessary we need to have an excellent understanding of the data data provided by the sensors :

Analyse values of the different gas to choose one :

- Calculate the gap between max and min values for each gas
- Compare values between sensors
- Proportion of missing value for each gas

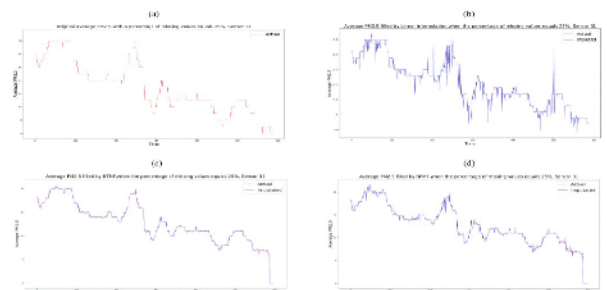
Determine a timestep where a sensor drift occur :

- Outliers : Difference in variation

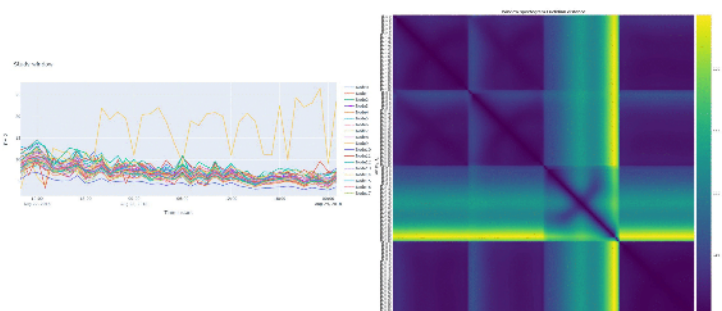
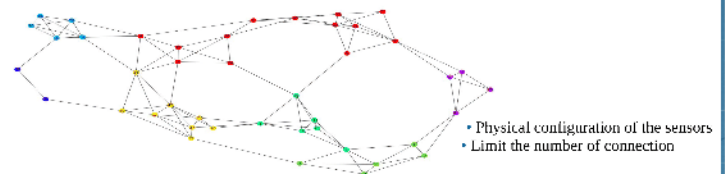


Data Imputation

- Homogenize the number of value per hour (calculate mean for each hour)
- Filling Nan Values using Bayesian Probabilistic Matrix Factorization technique



Network Implementation & Spectrogram



Conclusion :

- We show that it is possible to detect the concept drift from the distance matrix of the joint spectrograms. In this new representation the values with higher magnitude represent the outliers of the samples set.
- Thanks to the work carried out, it was demonstrated that GFT can be used for the spatio-temporal analysis of a sensor network.